

Notes: Jehiel and Weber (RES, 2026)

Endogenous Clustering and Analogy-Based Expectation Equilibrium

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1 Big picture

- **Question.** If agents cannot form fully game-specific expectations and instead compress many situations into a small number of categories, how should those categories themselves be chosen in equilibrium? And what kinds of strategic outcomes arise once categorization is endogenous rather than fixed from the outside?
- **Setting.** The paper studies a family of **normal-form two-player games** indexed by a state/game $\omega \in \Omega$. Players may observe many past behaviors across games, but due to cognitive or memory limits they sort games into at most K_i **analogy classes**. Within each class, they use a coarse expectation about the opponent's behavior.
- **Core challenge.** In the original analogy-based expectation equilibrium (ABEE) literature, analogy partitions are taken as **exogenous**. But if players choose categories to minimize prediction error, a fixed-point problem appears:
 - categories shape expectations,
 - expectations shape behavior,
 - behavior shapes which categories are prediction-optimal.

So categorization and play must be solved jointly.

- **Main conceptual contribution.** The paper endogenizes analogy partitions by importing ideas from **prototype theory** and **K-means clustering**. A clustered equilibrium requires:
 1. **given the partition**, strategies form an ABEE;
 2. **given the strategies**, the partition minimizes prediction errors, either locally or globally.
- **Why this is nontrivial.** Once the data to be clustered are endogenous, a pure steady state with one partition per player may fail to exist. The paper's first major message is that endogenous categorization can force **mixing over partitions**. That is not ordinary random noise; it is an equilibrium object.
- **Main theoretical results.**
 - The paper defines **Clustered Analogy-Based Expectation Equilibrium (CABEE)** and, when pure clustered equilibria may fail to exist, **Clustered Distributional ABEE (CD-ABEE)**.
 - In **finite environments**, there always exists at least one locally or globally clustered **distributional** ABEE when the divergence measure is either squared Euclidean distance or Kullback–Leibler divergence.
 - Some environments are **self-attractive**: once a partition is chosen, the induced behavior makes that same partition clustering-optimal again. This creates multiplicity.
 - Other environments are **self-repelling**: any candidate partition induces behavior that would reassign at least one game to another category. This creates equilibrium heterogeneity in analogy partitions.
- **Main applications.**
 - In a **beauty contest game**, stronger coordination motives make more and more partitions self-attractive; for r close to 1, virtually any partition can be sustained as part of a CABEE.

- In a **monitoring game**, any pure categorization is self-repelling, so steady state requires a mixture over partitions. This produces **polarized employer beliefs** about a minority worker type.
- In games with **linear best replies**, **strategic complements** tend to generate self-attractive partitions, while **strategic substitutes** generate self-repelling partitions.
- **Why it matters.** The paper gives a theory of how boundedly rational expectation formation can generate both:
 - **equilibrium multiplicity across societies** (different self-attractive categorizations), and
 - **belief heterogeneity within a society** (mixing over self-repelling categorizations).
- **Economic takeaway.** Categories are not a passive simplification device. In strategic settings they co-determine outcomes. Endogenous categorization can produce persistent disagreement, multiple stable modes of belief formation, and systematic departures from Nash behavior even when everyone sees the same objective data.

2 Data and measurement (key objects)

Because this is a **theory paper**, there is no empirical dataset in the usual sense. The relevant “data” are the **observed frequencies of opponents’ play across games**. The key measurement objects are therefore the game’s primitives, the analogy partitions, the representative behaviors, and the dispersion criteria used for clustering.

2.1 Strategic environment and primitives

- **Games.** There is a finite set of normal-form games indexed by $\omega \in \Omega$, with nature selecting game ω with probability $p(\omega)$.
- **Players.** The paper focuses on **two-player games**, $i = 1, 2$.
- **Actions.** In every game ω , player i has the same action set A_i .
- **Payoffs.** Player i ’s payoff is $u_i(a_i, a_j, \omega)$.
- **Strategies.** A strategy is $\sigma_i = (\sigma_i(\omega))_{\omega \in \Omega}$, where $\sigma_i(\omega) \in \Delta A_i$ can be mixed.
- **State observability.** Players observe which game ω they are in when they choose their action. The boundedness comes from coarse expectations, not from hidden states.

2.2 Analogy partitions and the memory bound

- **Analogy classes.** Player i can use at most K_i categories. An analogy partition is

$$An_i = \{\alpha_i^1, \dots, \alpha_i^{K_i}\},$$

a partition of Ω into K_i classes.

- **Interpretation of K_i .** The bound K_i represents a **memory or cognitive constraint**. Players can store only a small number of representative behaviors rather than a separate expectation for every game.
- **Category assignment.** For each game ω , $\alpha_i(\omega)$ denotes the unique analogy class to which ω belongs.
- **Representative behavior / prototype.** In each class α_i , player i forms an expectation $\beta_i(\alpha_i)$ about player j 's aggregate behavior in that class.

2.3 What gets clustered

- **Objects.** For player i , the objects to be clustered are the opponent's behaviors across games:

$$\{(\omega, \sigma_j(\omega))\}_{\omega \in \Omega}.$$

- **Important modeling choice.** Players cluster games based on the **distribution of opponent actions**, not on their own payoffs or on arbitrary state labels.
- **Reason.** If the goal of categorization is prediction, then the natural criterion is how well one category's representative behavior predicts the opponent's actual play across the games assigned to that category.

2.4 Distance, prediction error, and prototypes

- **Divergence measure.** The paper allows two benchmark measures of closeness between distributions:
 - squared Euclidean distance,
 - Kullback–Leibler divergence.
- **Prototype.** For any category $c \subseteq \Omega$, the prototype is the **mean behavior** in that category:

$$\beta_i(c) = \sum_{\omega \in c} p(\omega \mid c) \sigma_j(\omega).$$

- **Local clustering.** A game is assigned to the category with the nearest prototype.
- **Global clustering.** The partition minimizes total prediction error (variance or relative entropy depending on d).
- **Connection to machine learning.** Global clustering corresponds to the same objective as K-means-type procedures; local clustering corresponds to a local optimum condition.

2.5 Distribution over partitions and aggregate behavior

- **Why a distribution is needed.** Pure clustered equilibria may fail to exist. To restore existence, the paper lets each player use a **distribution over partitions**.
- **Notation.** λ_i is a probability distribution over the set of partitions $\mathcal{K}_i$.

- **Aggregate behavior.** Since different subjects may use different partitions, the opponent's observed behavior in a game is the average over partitions.
- **Interpretation.** The model no longer says everyone in a role must think in the same categories. Instead, heterogeneous categorizations can be a steady-state property of the environment.

2.6 Learning and timing

- **Learning model 1: raw data and machine learning.**
 - Stage 1: subjects see empirical frequencies of opponents' behavior by game and cluster them into K_i categories.
 - Stage 2: after learning which game applies, they best-respond to the representative behavior of that category.
- **Learning model 2: categories and cultural transmission.**
 - Subjects inherit categories and representative points from predecessors.
 - They play using inherited coarse expectations.
 - Afterwards, they re-cluster the newly generated data and pass the updated categories to successors.
- **Meaning of steady state.** In steady state, the distribution of categories and the induced strategic behavior reproduce each other.

3 Models and empirical specifications (all equations + interpretation)

3.1 Strategic environment and Nash benchmark

The expected payoff of player i in game ω when players use mixed actions $p_i \in \Delta A_i$ and $p_j \in \Delta A_j$ is

$$u_i(p_i, p_j, \omega) = \sum_{a_i, a_j} p_i(a_i) p_j(a_j) u_i(a_i, a_j, \omega).$$

A Nash equilibrium is a strategy profile $\sigma = (\sigma_i, \sigma_j)$ such that for every i , every game ω , and every alternative action distribution p_i ,

$$u_i(\sigma_i(\omega), \sigma_j(\omega), \omega) \geq u_i(p_i, \sigma_j(\omega), \omega).$$

Interpretation.

- Nash requires a game-specific correct forecast of the opponent's strategy in every ω .
- The whole point of the paper is to relax exactly this demanding requirement when there are many strategically distinct situations.

3.2 Analogy-Based Expectation Equilibrium (ABEE)

Given a partition An_i , player i forms an expectation $\beta_i(\alpha_i)$ in each analogy class. Consistency requires

$$\beta_i(\alpha_i) = \frac{\sum_{\omega \in \alpha_i} p(\omega) \sigma_j(\omega)}{\sum_{\omega \in \alpha_i} p(\omega)}.$$

Best response to coarse expectations requires

$$u_i(\sigma_i(\omega), \beta_i(\alpha_i(\omega)), \omega) \geq u_i(p_i, \beta_i(\alpha_i(\omega)), \omega)$$

for all $\omega \in \Omega$ and all $p_i \in \Delta A_i$.

So a strategy profile is an **ABEE** if:

1. each player's strategy is a best response to the representative behavior in the category that contains the current game;
2. those representative behaviors are the correct averages generated by the opponent's actual strategy.

Interpretation.

- ABEE already allows bounded rationality because players do not distinguish all games.
- But in standard ABEE, the partition is fixed in advance. The paper's new step is to make that partition endogenous.

3.3 Clustering: local and global

Player i compares how well different prototypes predict the opponent's play using a divergence d .

Local clustering. A partition An_i is locally clustered with respect to σ_j if for all classes α_i, α'_i in An_i and every $\omega \in \alpha_i$,

$$d(\sigma_j(\omega), \beta_i(\alpha_i)) \leq d(\sigma_j(\omega), \beta_i(\alpha'_i)).$$

Interpretation. Every game is assigned to the category whose representative behavior is closest to the opponent's actual behavior in that game.

Global clustering. A partition is globally clustered if it solves

$$An_i \in \arg \min_{P_i \in \mathcal{K}_i} \sum_{c_i \in P_i} p(c_i) \sum_{\omega \in c_i} p(\omega | c_i) d(\sigma_j(\omega), \beta_i(c_i)). \quad (1)$$

Interpretation.

- Under squared Euclidean distance, this objective is **variance minimization**.
- Under Kullback–Leibler divergence, it becomes **relative-entropy minimization**.
- Global clustering is stronger than local clustering; in the paper's setting, global clustering implies local clustering.

3.4 Clustered ABEE and the fixed-point problem

A pair (σ, An) is a **Clustered ABEE** if:

1. σ is an ABEE given An ;
2. for each player, An_i is locally or globally clustered with respect to the opponent's induced strategy.

Interpretation.

- This is the paper's core definition.
- The novelty is the fixed point: the data to be clustered are themselves endogenous because the strategy profile depends on the categorization.

3.5 Why pure clustered equilibria may fail: the matching-pennies example

The paper illustrates the problem with three matching-pennies games, $x \in \{a, b, c\}$, each with probability $1/3$, and payoff matrix

$$G_x = \begin{array}{c|cc} & L & R \\ \hline U & (1+x, 0) & (0, 1) \\ D & (0, 1) & (1, 0) \end{array}$$

where $0 < a < b < c < 2$.

Result (Proposition 1). If player 1 can use only $K_1 = 2$ categories and player 2 can use $K_2 = 3$, there is **no** clustered ABEE.

Interpretation.

- Bundling two matching-pennies games together changes how player 1 mixes.
- That change alters player 2's aggregate response in those games.
- Once the induced behavior is re-clustered, at least one game wants to move to a different category.
- So categorization is **self-repelling**: every candidate pure partition undoes itself.

This example motivates the distributional extension.

3.6 Clustered Distributional ABEE (CD-ABEE)

Let λ_i be a distribution over partitions in $\mathcal{K}_i$. If different subjects in role j use different partitions, the **aggregate strategy** in game ω is

$$\bar{\sigma}_j(\omega) = \sum_{An_j \in \mathcal{K}_j} \lambda_j(An_j) \sigma_j(\omega \mid An_j). \quad (2)$$

Consistency of expectations within a chosen partition An_i becomes

$$\beta_i(\alpha_i \mid An_i) = \frac{\sum_{\omega \in \alpha_i} p(\omega) \bar{\sigma}_j(\omega)}{\sum_{\omega \in \alpha_i} p(\omega)}. \quad (3)$$

A **distributional ABEE** requires best responses and consistency for every partition in the support of λ_i . A **clustered distributional ABEE** adds the requirement that every partition in the support is locally or globally clustered with respect to the aggregate opponent strategy.

Existence theorem.

Theorem 1. In finite environments, there always exists a locally (and globally) clustered distributional ABEE when the divergence is squared Euclidean distance or Kullback–Leibler divergence.

Interpretation.

- Mixing over partitions is not a technical nuisance. It is what restores existence when pure clustered equilibria fail.
- In equilibrium, different agents exposed to the same raw evidence may categorize the world differently.

3.7 Learning foundations

Learning model 1: raw data and machine learning.

- Subjects first observe game-by-game frequencies of opponents' past actions.
- They cluster those raw data into K_i categories, possibly via a full clustering solution or a K-means-type local search.
- After learning which game applies, they best-respond to the representative behavior of the selected category.

Learning model 2: categories and cultural transmission.

- Subjects inherit categories and representative points from predecessors.
- They play using those inherited coarse expectations.
- After observing newly generated data, they update categories and pass them on.

Interpretation.

- Learning model 1 provides a foundation for **globally clustered distributional ABEE**.
- Learning model 2 provides a foundation for **locally clustered distributional ABEE**.
- The lag between seeing raw data and learning which game applies is crucial: constrained memory transmits compressed information from the first stage to the second.

3.8 Application 1: beauty contest and self-attractive partitions

A state/fundamental $\theta \in \Theta \subset \mathbb{R}$ is drawn from a smooth density. Player i chooses $a_i \in \mathbb{R}$ and has utility

$$U_i(a_i, a_{-i}; \theta) = -(1-r)(a_i - \theta)^2 - r(a_i - a_j)^2, \quad 0 < r < 1.$$

The best response to an expected opponent strategy σ_j is

$$\text{BR}(\theta, \sigma_j) = (1-r)\theta + r \mathbb{E}(a_j \mid \sigma_j).$$

The unique Nash equilibrium is

$$a_1^{NE}(\theta) = a_2^{NE}(\theta) = \theta.$$

Now suppose both players use the same partition $(\Theta_k)_{k=1}^K$, with class means

$$\bar{\theta}_k = \mathbb{E}(\theta \mid \theta \in \Theta_k).$$

Then the unique ABEE is

$$a_1^{ABEE}(\theta) = a_2^{ABEE}(\theta) = (1 - r)\theta + r\bar{\theta}_k, \quad \theta \in \Theta_k. \quad (4)$$

Interpretation.

- Players no longer respond exactly to the fundamental.
- Inside each category, behavior is pulled toward the category mean $\bar{\theta}_k$.
- The larger is r , the more strongly behavior is compressed toward the category prototype.

Main result (Proposition 2). For any partition whose class means are distinct, if r is sufficiently close to 1, that partition together with the induced ABEE is a CABEE.

Monotonicity result (Proposition 3). If a partition is a locally clustered CABEE at some $r < 1$, then it remains a locally clustered CABEE for any larger $r' > r$.

Interpretation.

- Stronger coordination motives make within-class behavior bunch more tightly around class means.
- That makes the originally chosen partition more likely to be re-selected by clustering.
- Hence beauty-contest environments are **self-attractive**: many different categorizations can become self-confirming.

3.9 Application 2: monitoring game and self-repelling partitions

There are three worker types a, b, c , observed by the employer. The employer chooses whether to **Control** (C) or not (D); the worker chooses effort $e \in \{0, 1\}$.

Assumptions:

- Type a always chooses low effort $e = 0$.
- Type b always chooses high effort $e = 1$.
- Type c chooses low effort if and only if the probability of control is no larger than μ^* .
- The employer prefers control only if she expects low effort with probability at least ν^* .

In the unique Nash equilibrium:

- the employer chooses C against type a ,
- the employer chooses D against type b ,
- against type c , the employer mixes with probability μ^* on C ,

- type c mixes with probability ν^* on low effort.

Assume the employer can use only $K = 2$ categories.

Main result (Proposition 4). Under suitable probability restrictions, there is **no pure CABEE** with a single employer partition.

Global distributional result (Proposition 5). There is a unique globally clustered distributional ABEE in which the employer mixes between the partitions that group c with a and with b .

Local distributional result (Proposition 6). Under symmetry assumptions, there is a whole interval of locally clustered distributional equilibria, and the admissible range of mixing is wider under KL divergence than under squared Euclidean distance.

Interpretation.

- This environment is **self-repelling**: no single pure categorization reproduces itself.
- Steady state therefore requires **heterogeneous employer categorizations**.
- Some employers group the minority type c with low-effort workers and over-control them; others group them with high-effort workers and under-control them.
- The model therefore offers a new mechanism for **polarized beliefs and differential treatment of minorities** even when everyone faces the same objective data.

3.10 Application 3: linear best replies

Consider games indexed by an interaction parameter μ . Player i 's best response is

$$\text{BR}_i(\mu, \sigma_j) = A + \mu B + \mu C \mathbb{E}_{\sigma_j}(a_j), \quad 0 < C < 1.$$

This nests familiar environments such as differentiated-product Bertrand or Cournot competition with linear demands.

The unique Nash equilibrium is

$$a_1^{NE}(\mu) = a_2^{NE}(\mu) = \frac{A + \mu B}{1 - \mu C}.$$

Assume players use a common **interval analogy partition**

$$An_1 = An_2 = \{[\mu_0, \mu_1], (\mu_1, \mu_2], \dots, (\mu_{K-1}, \mu_K]\}.$$

Unique ABEE under a given interval partition (Proposition 7). For each class α_k , the representative expectation is

$$\beta(\alpha_k) = \frac{A + B \mathbb{E}[\mu \mid \alpha_k]}{1 - C \mathbb{E}[\mu \mid \alpha_k]},$$

and the induced action in game $\mu \in \alpha_k$ is

$$a(\mu \mid \alpha_k) = A + \mu \frac{B + AC}{1 - C \mathbb{E}[\mu \mid \alpha_k]}.$$

Interpretation.

- The class mean of μ determines the prototype expectation $\beta(\alpha_k)$.
- Once that class mean is fixed, actions inside the class are linear in μ .
- Whether local clustering succeeds depends on how the jumps between classes interact with the slope inside each class.

3.11 Strategic complements: self-attraction

Take $\mu \in [0, 1]$. Then the induced ABEE function has discontinuities at the class boundaries. The key feature is that the jumps go in the **same direction** as the slope inside each class.

A convenient local-clustering condition is

$$a(\mu_k \mid \alpha_k) \leq \frac{\beta(\alpha_k) + \beta(\alpha_{k+1})}{2} \leq a(\mu_k \mid \alpha_{k+1}).$$

The paper then introduces an **equidistant-expectations partition**: boundaries are chosen so that each boundary point μ_k is equally far from the means of the neighboring classes.

Main result (Proposition 8). Around any equidistant-expectations partition there exists an open set of nearby interval boundaries that support a locally clustered ABEE.

Interpretation.

- Strategic complements make the induced ABEE points within a class look **closer together** than under no interaction.
- This creates slack in the local-clustering inequalities.
- The stronger is complementarity, the more partitions become self-attractive.

3.12 Strategic substitutes: self-repulsion

Now take $\mu \in [-1, 0]$. The crucial difference is that the induced ABEE function becomes **non-monotonic**: the jumps at class boundaries go in the **opposite direction** from the within-class slope.

Main result (Proposition 9). Whenever $B \neq -AC$, there are no symmetric interval analogy partitions that are locally clustered with respect to the induced ABEE.

Interpretation.

- The non-monotonicity creates local reassignment pressure around every class boundary.
- Hence strategic-substitute environments are naturally **self-repelling**.
- The paper's working-paper extension studies the associated distributional equilibrium when μ takes only three values.

4 Identification and instruments (what makes the estimates credible)

This paper is theoretical, so there are no empirical instruments. The relevant question is instead: *what makes the conceptual and mathematical results persuasive?*

4.1 What fails in the standard ABEE approach

- Standard ABEE takes analogy partitions as given.
- But if players choose categories to improve prediction, exogenous partitions are not satisfactory.
- Once the partition changes, the induced data change too, because play changes. This creates an endogenous-data problem absent from standard clustering and absent from standard ABEE.

4.2 Why the clustering formulation is disciplined

- The paper does not let categories drift arbitrarily. It ties them to a precise **prediction-error minimization** problem.
- The prototype in each class is the **mean behavior** within that class, which is optimal for the divergence measures considered.
- The use of squared Euclidean and KL divergences connects the theory directly to standard tools in statistics, information theory, and machine learning.
- The local/global distinction is important:
 - local clustering corresponds to “nearest prototype” assignment,
 - global clustering corresponds to full dispersion minimization.

4.3 Why the distributional extension is essential

- The matching-pennies example shows that **pure** clustered equilibria may fail to exist.
- The move to distributions over partitions is therefore not cosmetic. It is the correct analogue of mixed equilibrium when pure self-consistency of categories is impossible.
- This also changes the interpretation of equilibrium: heterogeneity in categorization can be an **equilibrium necessity**, not a modeling convenience.

4.4 Why the learning foundations matter

- The two learning models explain how these equilibrium concepts could arise from decentralized adaptation rather than from direct fixed-point reasoning alone.
- Learning model 1 gives a natural “machine learning” interpretation: agents compress raw historical data before acting.
- Learning model 2 gives a “cultural transmission” interpretation: categories are inherited, used, and then updated.
- These foundations clarify why clustered distributional equilibria can describe socially persistent modes of expectation formation.

4.5 Relation to the literature

- **Behavioral game theory.** The paper belongs to the literature relaxing rational expectations and fully specified beliefs.
- **Analogy-based expectations.** It extends Jehiel’s ABEE approach by endogenizing the partitions.
- **Psychology and categorization.** It borrows prototype-style reasoning from psychology.
- **Machine learning.** It connects equilibrium reasoning to K-means clustering and local/global minimization.
- **Self-confirming and misspecified equilibria.** The paper is close in spirit but differs because the coarse statistics are **endogenously generated by categorization** rather than imposed exogenously.

4.6 What the paper’s applications establish

- Beauty-contest environments show how endogenous categories can create **many self-confirming equilibria**.
- Monitoring environments show how endogenous categories can force **mixtures over beliefs** and thereby produce polarized treatment.
- Linear best-reply environments isolate a clean structural distinction:
 - strategic complements $\Rightarrow$ self-attraction,
 - strategic substitutes $\Rightarrow$ self-repulsion.

5 Tables and figures

This is a **theory-driven paper**: there are no empirical tables. The main visual evidence comes from one illustrative example and two figures that summarize the geometry of local clustering.

5.1 Example 1 and Proposition 1: matching pennies

- This example is the paper’s first proof that pure clustered equilibria may fail to exist.
- The payoff asymmetry across the three matching-pennies games makes the row player’s optimal mixing sensitive to how games are bundled.
- Once two games are put in the same category, induced play becomes polarized enough that clustering wants to reshuffle one of them.
- Read this example as the motivation for **CD-ABEE**. Without it, the distributional extension would look like an abstract technical refinement; with it, it becomes economically necessary.

5.2 Figure 1: strategic complements

- Figure 1 plots Nash and ABEE action schedules under **strategic complements**.
- The key visual fact is that the ABEE schedule has **jumps in the same direction as the within-class slope**. This preserves the geometric possibility of local clustering.
- Read the figure as showing why complementarity makes categories **self-attractive**. Once a partition is chosen, the resulting actions bunch in a way that helps justify that same partition.
- The more important lesson is not the exact parameter values used in the illustration, but the qualitative shape: monotonic segments plus same-direction jumps.

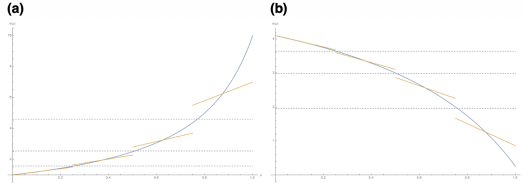


FIGURE 1
Local clustering for $K=4$ and μ -sequence such that $\mu_k = \frac{1}{4}, k=0, 1, \dots, 4$. (a) Increasing NE and ABEE functions with $A=1, B=1$ and $C=0.8$. (b) Decreasing NE and ABEE functions with $A=4.1, B=-4$ and $C=0.6$

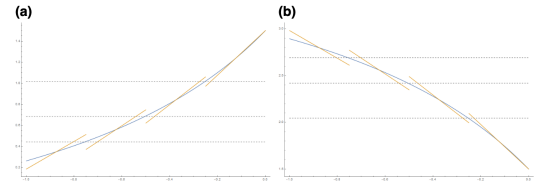


FIGURE 2
Local clustering for $K=4$ and μ -sequence such that $\mu_k = \frac{3}{4}, k=0, 1, \dots, 4$. (a) Increasing NE and ABEE functions with $A=1.5, B=1$, and $C=0.9$. (b) Decreasing NE and ABEE functions with $A=1.5, B=-4$, and $C=0.9$

5.3 Figure 2: strategic substitutes

- Figure 2 plots the corresponding schedules under **strategic substitutes**.
- Now the jumps go in the **opposite direction** from the within-class slope, which creates local non-monotonicity around category boundaries.
- This visualizes why local clustering fails: points just on the two sides of a boundary cannot simultaneously be closer to their own class prototypes.
- Read the figure as the graphical explanation for Proposition 9 and for the broader claim that strategic-substitute environments are **self-repelling**.

6 Short summary

- **Method.** Start from ABEE, then endogenize the analogy partition by requiring that categories minimize prediction errors about the opponent's behavior.
- **Main conceptual move.** Because behavior depends on categories and categories depend on behavior, equilibrium must solve a fixed point between play and clustering.
- **Existence.** Pure clustered equilibria may fail, but clustered **distributional** ABEE always exists in finite environments.
- **Learning.** The equilibrium concepts arise naturally from two-stage learning models based on raw-data clustering or inherited categories.
- **Applications.** Beauty contest games generate self-attractive partitions and multiplicity; monitoring games generate self-repelling partitions and heterogeneous beliefs; linear best-reply games show that complements and substitutes behave very differently.

- **Main result.** Endogenous categorization can itself be a source of equilibrium multiplicity, disagreement, and persistent departures from the fully rational benchmark.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes a new equilibrium concept in which **coarse expectation formation and endogenous categorization are jointly determined**.
- It shows that when categories are chosen to minimize prediction error, equilibrium may require either:
 - multiple self-consistent partitions, or
 - mixtures over partitions.
- Therefore, bounded rationality in strategic settings is not only about using coarse beliefs. It is also about how those coarse beliefs are **organized**. That organizational choice has first-order strategic consequences.

7.2 Economic intuition

- If agents compress many situations into a few categories, then they inevitably respond to averages rather than to fully specific states.
- But once everyone does this, the averages they respond to are themselves shaped by the chosen categorization.
- In environments with strategic complements, behaving similarly inside a category makes that category more defensible ex post.
- In environments with strategic substitutes, the induced behavior tends to undermine the original grouping, so the only way to maintain a steady state is through heterogeneity in categorization across agents.

7.3 Takeaway

This paper's central lesson is that **how people partition the world is an equilibrium object**. Once categories are chosen endogenously, the economy can support multiple stable ways of thinking, or require persistent heterogeneity in how identical evidence is processed. That makes endogenous categorization a fundamental new source of both **equilibrium multiplicity** and **belief polarization**.